Discussion 08

Material Modeling

Computer Graphics and Imaging
UC Berkeley CS 184/284A

Categories of Reflection Functions

Ideal specular

Ideal specular

- **Perfect mirror reflection**

Ideal diffuse

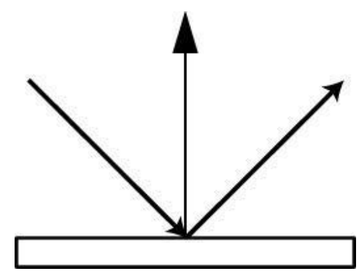
- **Equal reflection in all directions**

Glossy specular

- **Majority of light reflected near mirror direction**

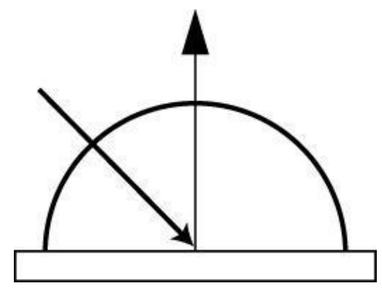
Retro-reflective

- **Light reflected back towards light source**

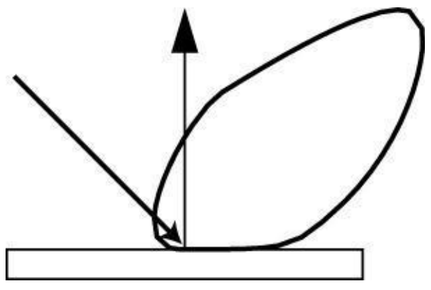






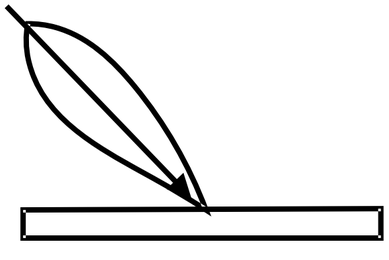


















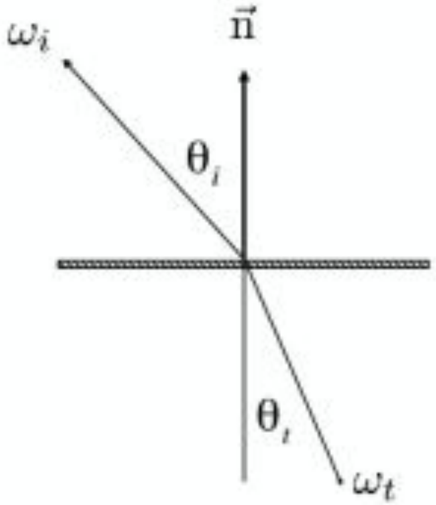
Diagrams illustrate how light from incoming direction is reflected in various outgoing directions.

Snell's Law of Refraction

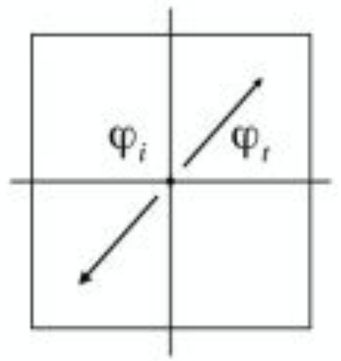
Light bends when it crosses a boundary between two media with different indices of refraction (IOR).

$$\eta_i \sin(\theta_i) = \eta_t \sin(\theta_t)$$

- η_i, η_t: IOR of incident, transmitted media
- θ_i, θ_t: Angles to the surface normal



$$\eta_i \sin\theta_i = \eta_t \sin\theta_t$$



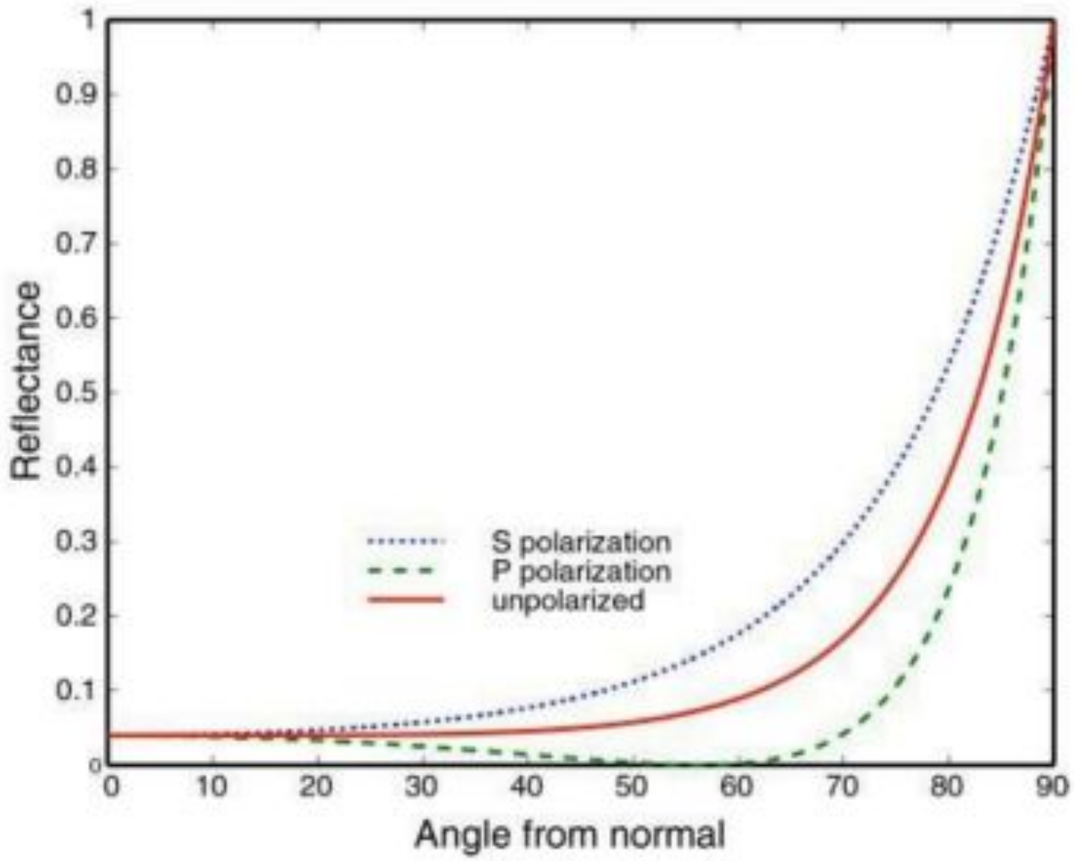
$$\varphi_t = \varphi_i \pm \pi$$

Medium	η^*
Vacuum	1.0
Air (sea level)	1.00029
Water (20°C)	1.333
Glass	1.5-1.6
Diamond	2.42

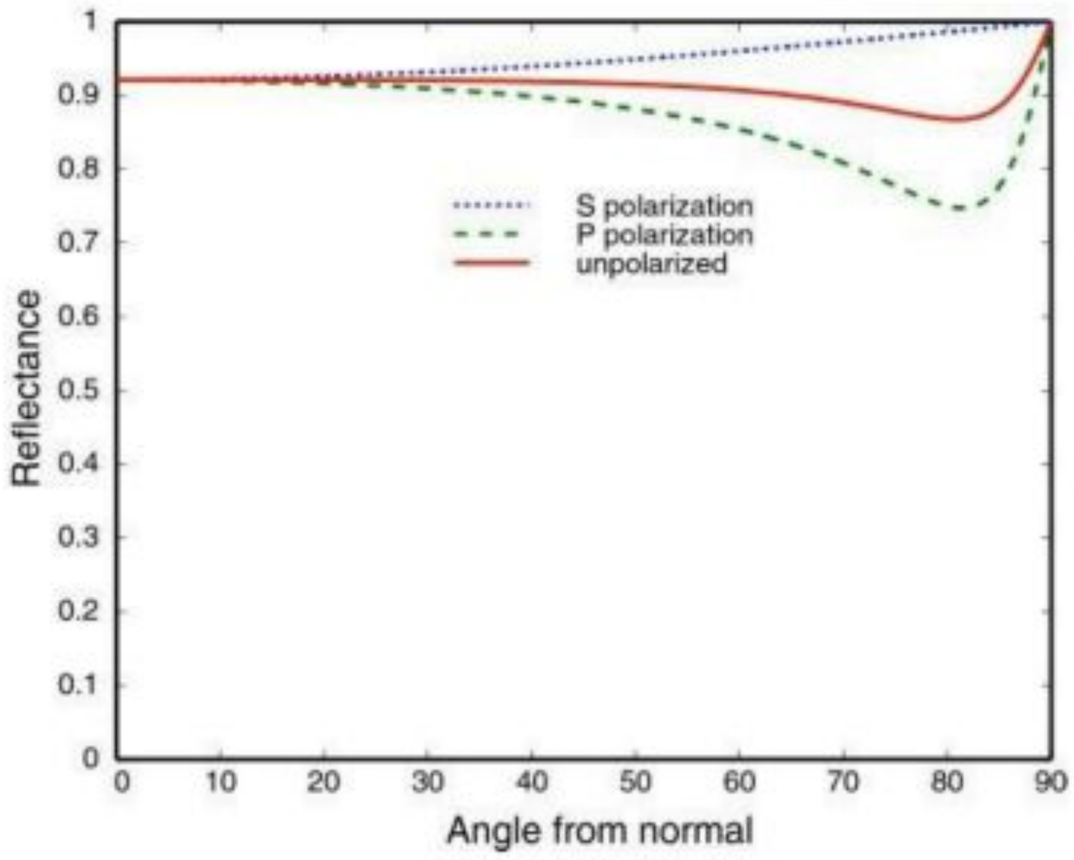
* index of refraction is wavelength dependent (these are averages)

The Fresnel Term (n*)

The Fresnel equations describe how a surface's reflectivity changes with viewing angle and material type. They determine the ratio of reflected to transmitted light.



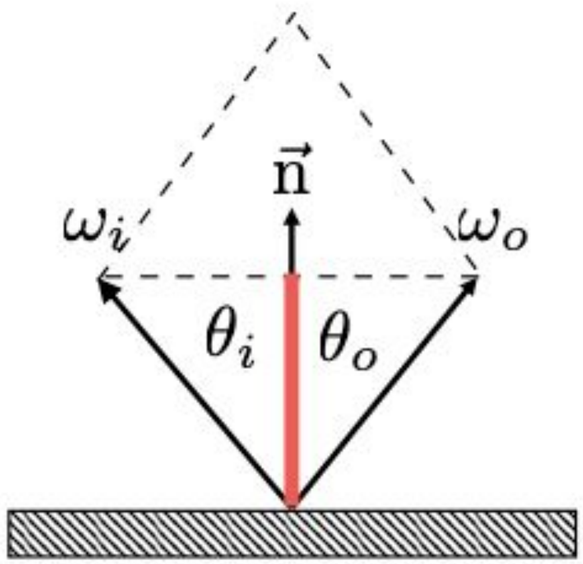
Dielectrics (e.g., plastic, water) reflect very little head-on, but much more at grazing angles.



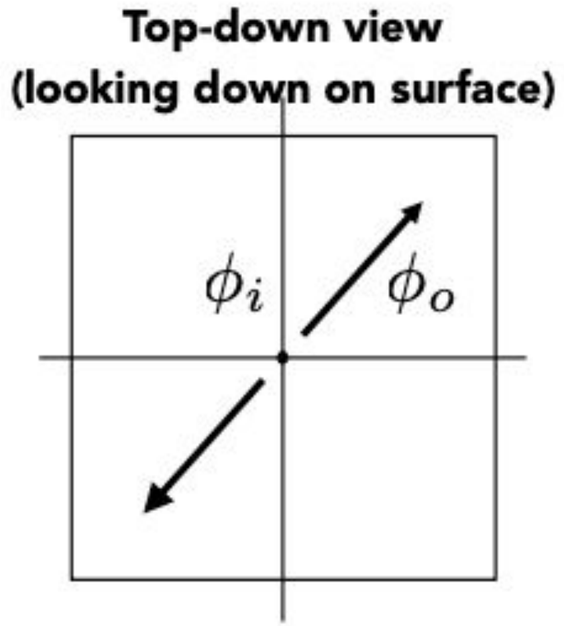
Conductors (e.g., metals) are highly reflective at all angles and have a colored reflectance.

Perfect Specular Reflection:

How to compute bounce



$$\theta = \theta_o = \theta_i$$



$$\phi_o = (\phi_i + \pi) \bmod 2\pi$$

$$\omega_o + \omega_i = 2\cos\theta\,\vec{n} = 2(\omega_i \cdot \vec{n})\vec{n}$$

$$\omega_o = -\omega_i + 2(\omega_i \cdot \vec{n})\vec{n}$$

Total Internal Reflection (TIR)

When light goes from a dense to a less dense medium ($\eta_i > \eta_t$), if θ_i is greater than the **critical angle** θ_c, all light is reflected, and none is refracted.

TIR occurs when Snell's law would require $\sin(\theta_t) > 1$, which is impossible. The boundary case $\sin(\theta_t) = 1$ (i.e., $\theta_t = 90^\circ$) gives the critical angle:

$$\eta_i \sin(\theta_c) = \eta_t \sin(90^\circ) = \eta_t$$

$$\theta_c = \arcsin\left(\frac{\eta_t}{\eta_i}\right)$$





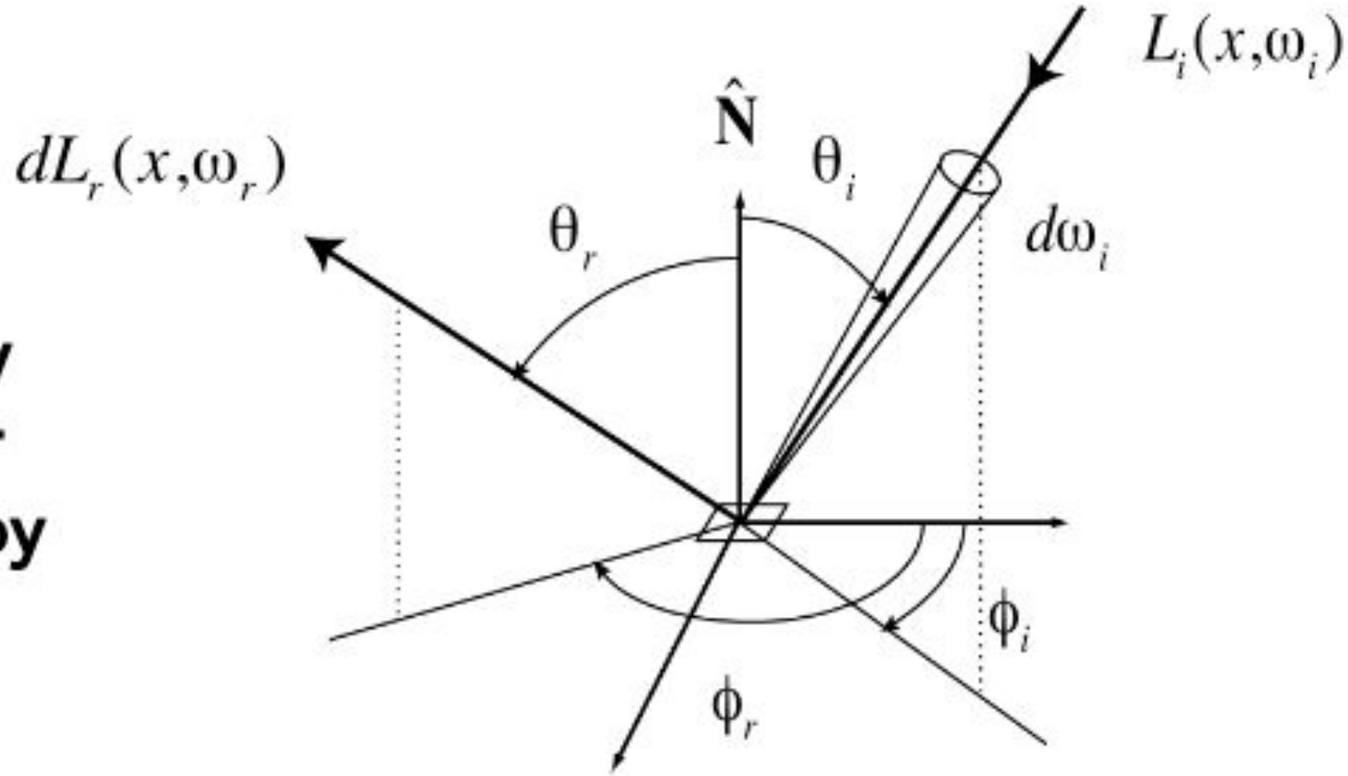
Snell's Window is a result of TIR.

BRDF

The bidirectional reflectance distribution function

BRDF

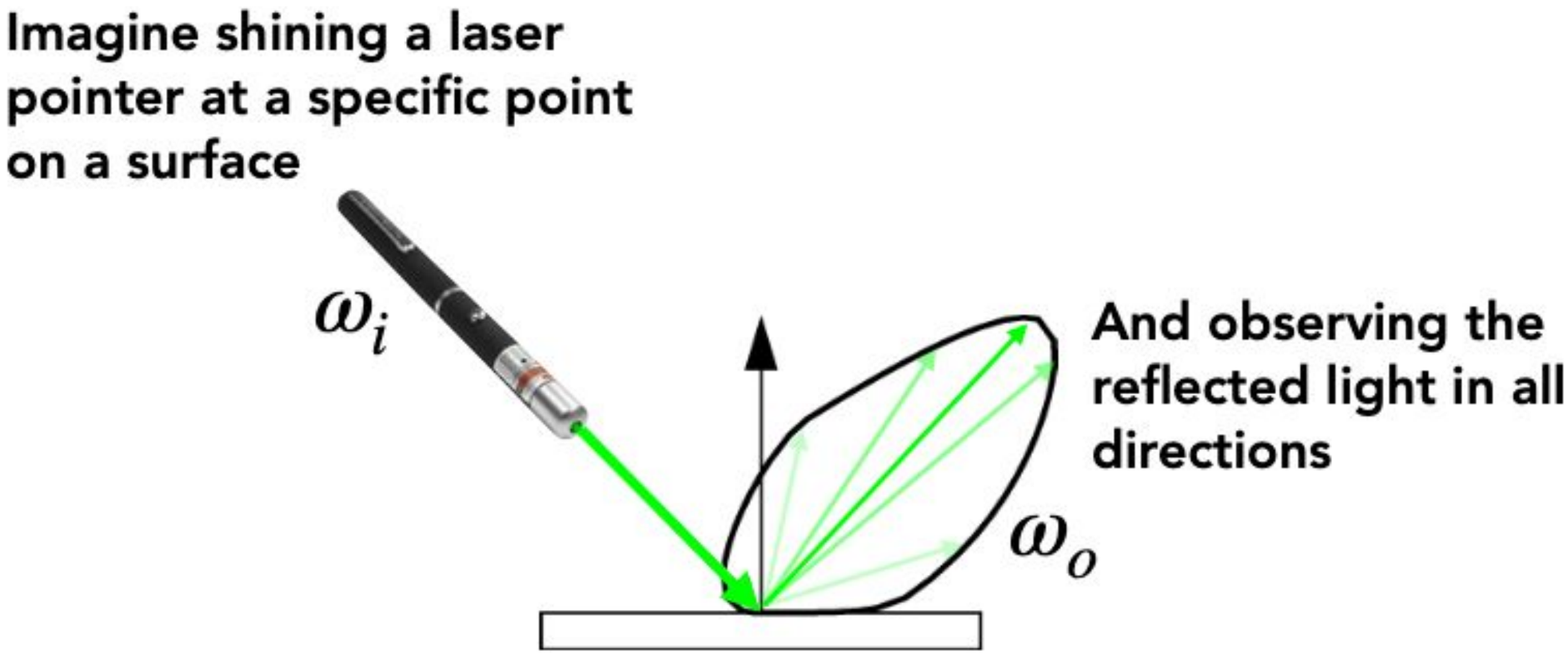
Definition: The bidirectional reflectance distribution function (BRDF) represents how much light is reflected into each outgoing direction ω_r from each incoming direction



NB: ω_i points away from surface rather than into surface, by convention.

$$f_r(\omega_i \to \omega_r) = \frac{\mathrm{d}L_r(\omega_r)}{\mathrm{d}E_i(\omega_i)} = \frac{\mathrm{d}L_r(\omega_r)}{L_i(\omega_i)\cos\theta_i\,\mathrm{d}\omega_i} \quad \left[\frac{1}{\mathrm{sr}}\right]$$

What is a BRDF?



Important properties

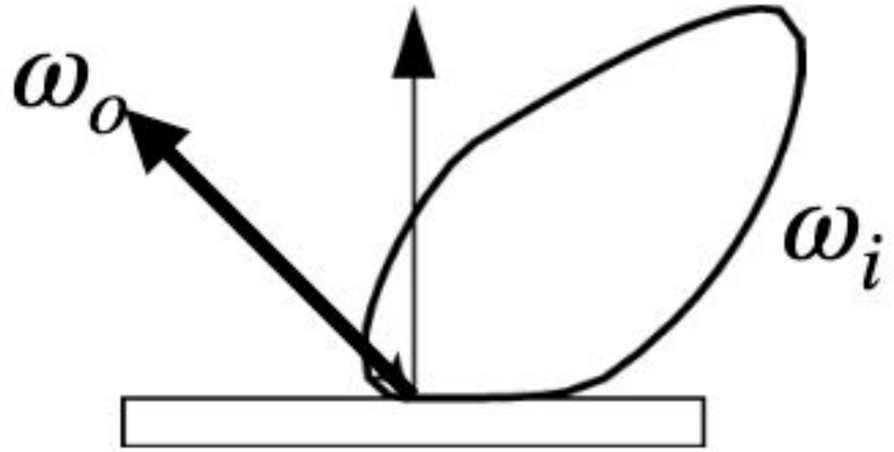
Positivity (self explanatory)

$$f(\omega_i, \omega_o) \geq 0$$

Important properties

- **Reciprocity (can trace light paths either direction)**

$$f(\omega_i, \omega_o) = f(\omega_o, \omega_i)$$

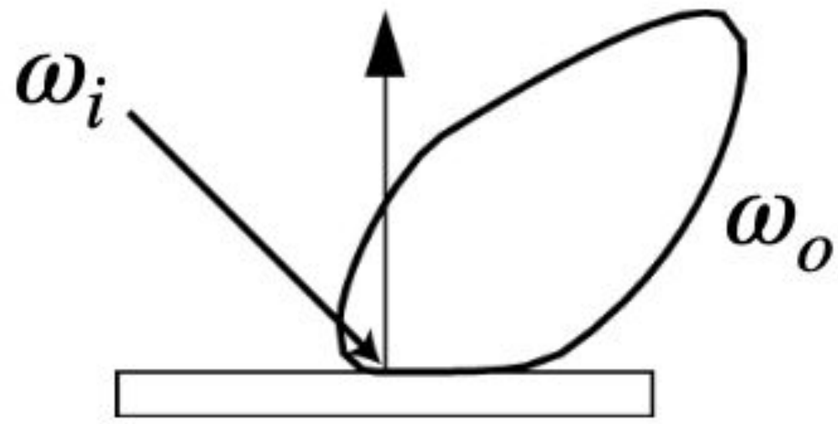


Means you can swap the labels in this diagram

Important properties

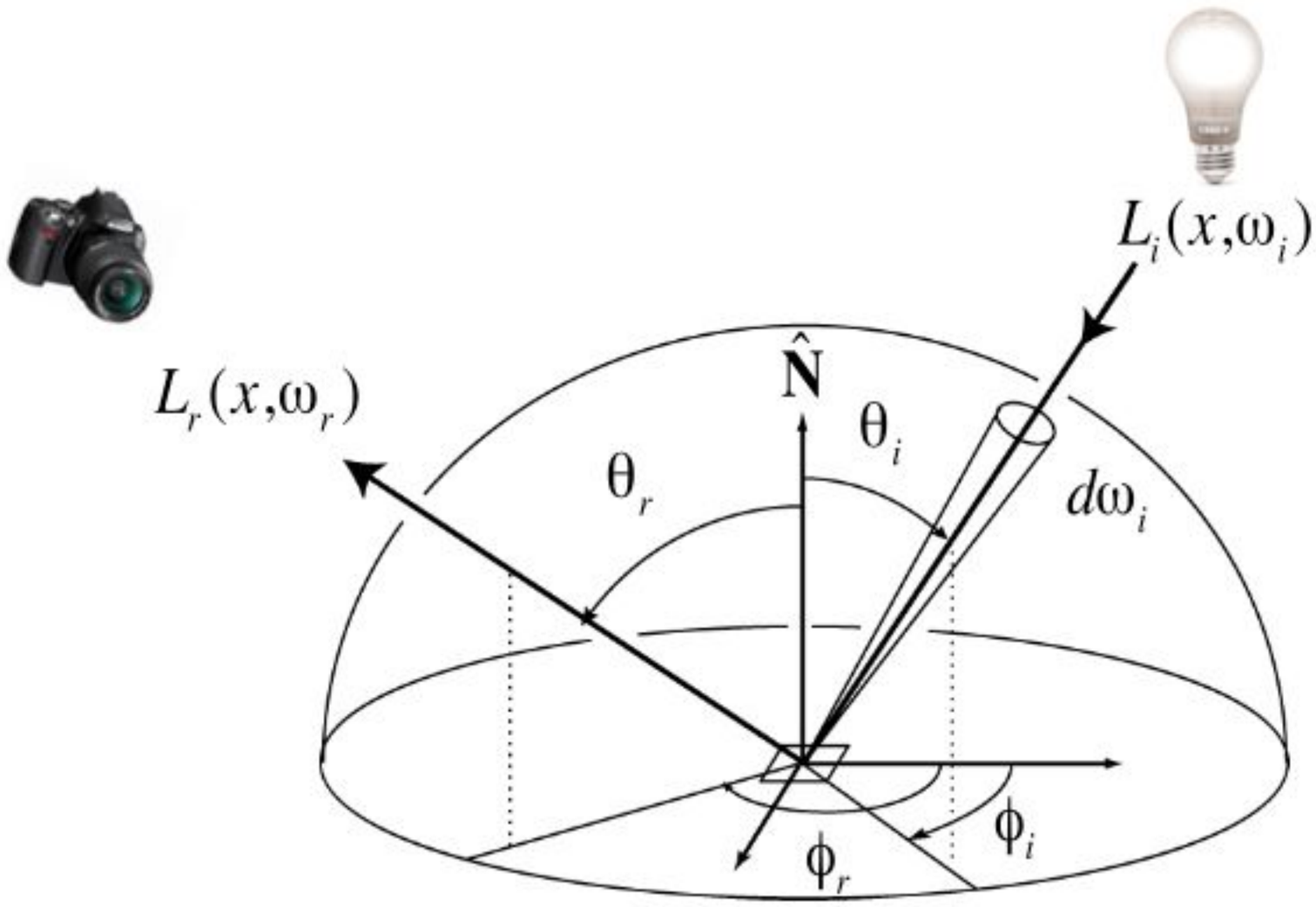
- Conservation of energy (can't reflect > 100%)

$$\int_{\Omega} f(\omega_i, \omega_o) \cos\theta_o d\omega_o \leq 1$$



If you add up all the radiance in the "lobe" it can't exceed 1, given 1 unit of input radiance

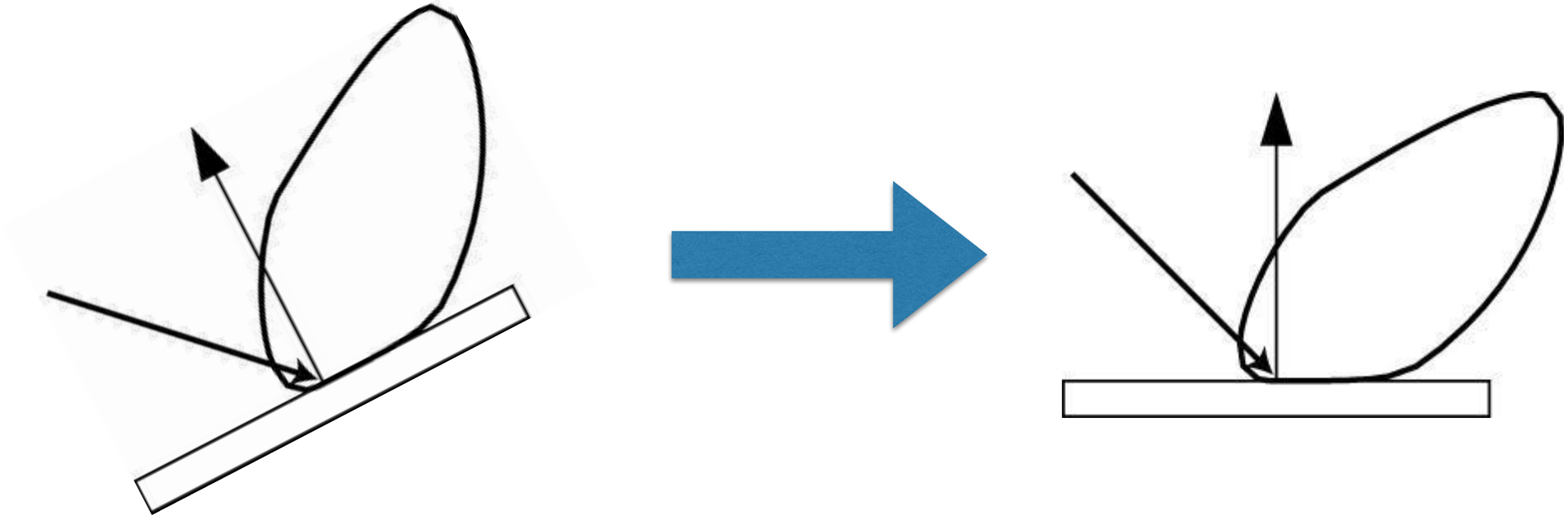
How you use a BRDF: reflection equation



$$L_r(\mathrm{p},\omega_r) = \int_{H^2} f_r(\mathrm{p},\omega_i \rightarrow \omega_r)\, L_i(\mathrm{p},\omega_i)\, \cos\theta_i\, \mathrm{d}\omega_i$$

BRDF “coordinate system” in assignment

- Align the normal with the z axis (0,0,1)
- This simplifies BRDF evaluation math
- ONLY valid for a single point on an object!
- Not the same as “object space”

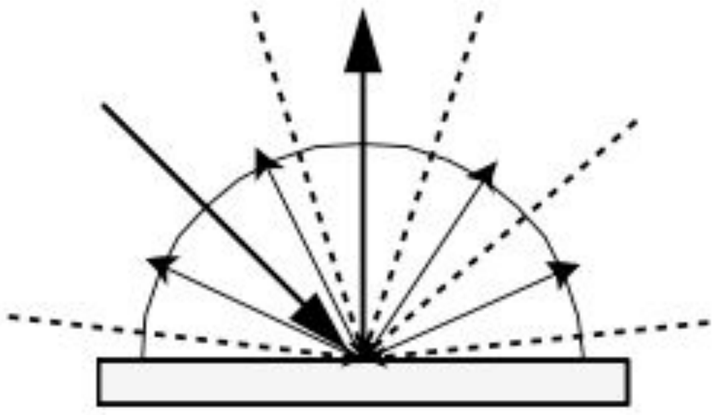


World-space orientation of surface

Local BRDF-space defined by normal vector

Diffuse / Lambertian Material

Light is equally reflected in each output direction



$$f_r = c$$

Suppose the incident lighting is uniform:

$$L_o(\omega_o) = \int_{H^2} f_r L_i(\omega_i) \cos\theta_i \, \mathrm{d}\omega_i$$

$$= f_r L_i \int_{H^2} \cancel{(\omega_i)} \cos\theta_i \, \mathrm{d}\omega_i$$

$$= \pi f_r L_i$$

$$f_r = \frac{\rho}{\pi}$$ — albedo (color)

Work Problem 1: BRDF Physical Plausibility

Worksheet Section 2, Problem 6 - ~10 minutes

6. **BRDF Physical Plausibility**

   A physically plausible BRDF must satisfy certain properties. Consider the following BRDF:

$$f_r(\omega_i, \omega_o) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot n)\,(\omega_o \cdot n)}$$

   where k and n are constants, ω_i is the incoming direction, ω_o is the outgoing direction, and n is the surface normal.

   1. Check if this BRDF satisfies positivity.
   2. Check if this BRDF satisfies Helmholtz reciprocity (symmetry).
   3. Find the constraints on k and n that would make this BRDF conserve energy.

Solution 1: Positivity & Reciprocity

Worksheet Section 2, Problem 6

Solution:

1. Positivity requires that $f_r(\omega_i, \omega_o) \geq 0$ for all valid ω_i and ω_o.

   For physically valid directions (above the surface), we have $(\omega_i \cdot n) \geq 0$ and $(\omega_o \cdot n) \geq 0$.

   If $k \geq 0$ and n is even, then $(\omega_i \cdot \omega_o)^n \geq 0$ for all directions.

   If n is odd, then $(\omega_i \cdot \omega_o)^n$ can be negative when $(\omega_i \cdot \omega_o) < 0$.

   Therefore, the BRDF satisfies positivity if $k \geq 0$ and n is even.

2. Helmholtz reciprocity requires that $f_r(\omega_i, \omega_o) = f_r(\omega_o, \omega_i)$.

   Let's check:

$$f_r(\omega_i, \omega_o) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot n)\,(\omega_o \cdot n)}$$

   Swapping ω_i and ω_o:

$$f_r(\omega_o, \omega_i) = \frac{k\,(\omega_o \cdot \omega_i)^n}{(\omega_o \cdot n)\,(\omega_i \cdot n)}$$

   Since the dot product is commutative, $(\omega_i \cdot \omega_o) = (\omega_o \cdot \omega_i)$. Therefore:

$$f_r(\omega_o, \omega_i) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_o \cdot n)\,(\omega_i \cdot n)} = f_r(\omega_i, \omega_o)$$

   So the BRDF satisfies Helmholtz reciprocity.

Solution 1: Energy Conservation

Worksheet Section 2, Problem 6

3. Energy conservation requires that for any incoming direction ω_i, the total energy reflected in all directions cannot exceed the energy received:

$$\int_\Omega f_r(\omega_i, \omega_o)\,(\omega_o \cdot n)\,d\omega_o \leq 1$$

Substituting our BRDF:

$$\int_\Omega \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot n)\,(\omega_o \cdot n)}\,(\omega_o \cdot n)\,d\omega_o = \frac{k}{(\omega_i \cdot n)}\int_\Omega (\omega_i \cdot \omega_o)^n\,d\omega_o$$

This integral depends on the value of n. For energy conservation, we need:

$$\frac{k}{(\omega_i \cdot n)}\int_\Omega (\omega_i \cdot \omega_o)^n\,d\omega_o \leq 1 \text{ for all } \omega_i$$

The worst case is when $\omega_i = n$ (normal incidence), which maximizes the integral. In this case:

$$k\int_\Omega (\cos\theta_o)^n\,d\omega_o \leq 1$$

The integral evaluates to $\frac{2\pi}{n+1}$ for even n. Therefore:

$$k \leq \frac{n+1}{2\pi}$$

This is the constraint on k for energy conservation, assuming n is even (which we need for positivity).

Microfacet BRDFs

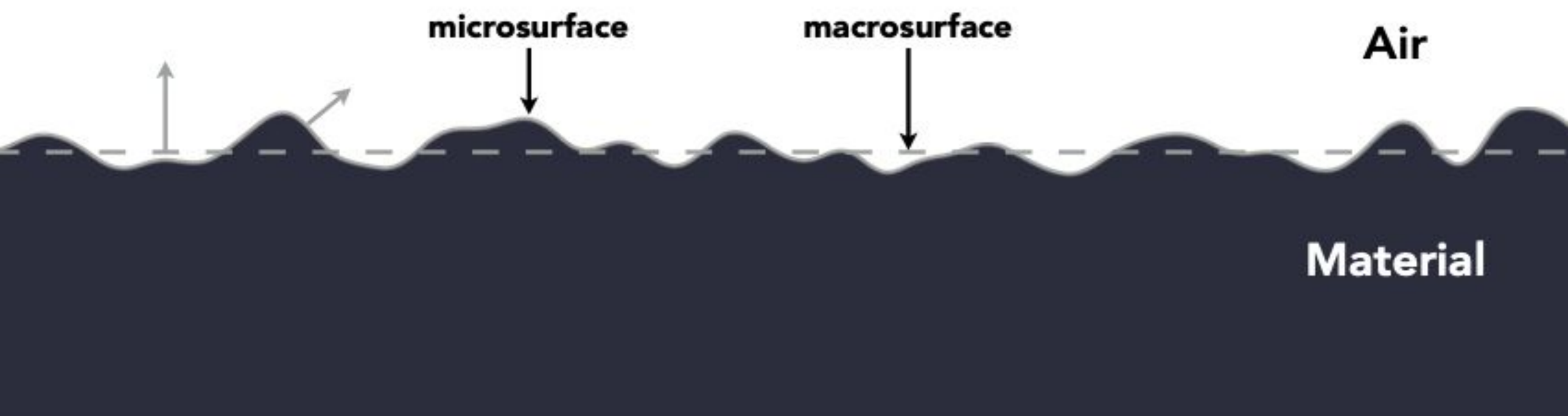
Microfacet Theory

Rough surface

- **Macroscale: flat & rough**
- **Microscale: bumpy & specular**

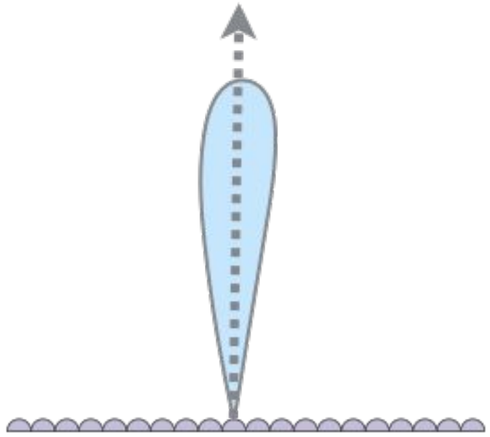
Individual elements of surface act like mirrors

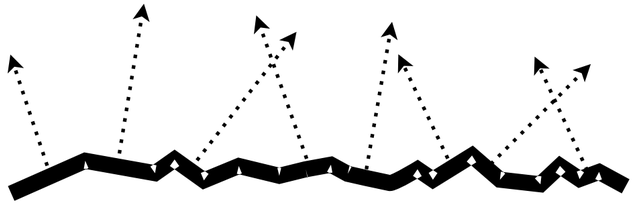
- **Known as Microfacets**
- **Each microfacet has its own normal**

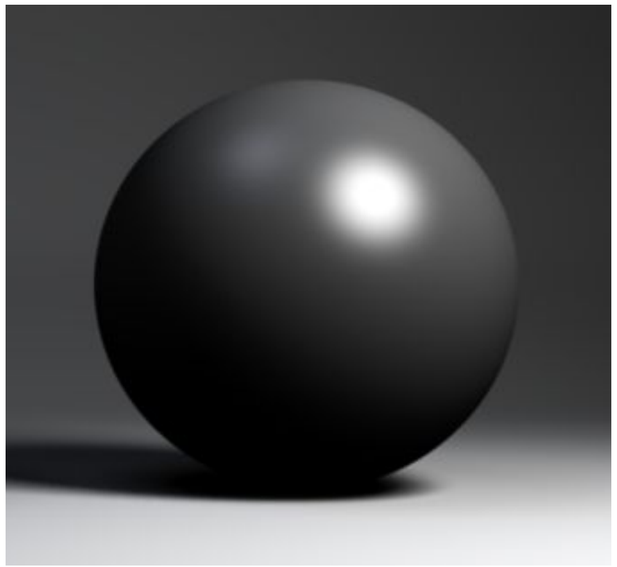


Microfacet BRDF

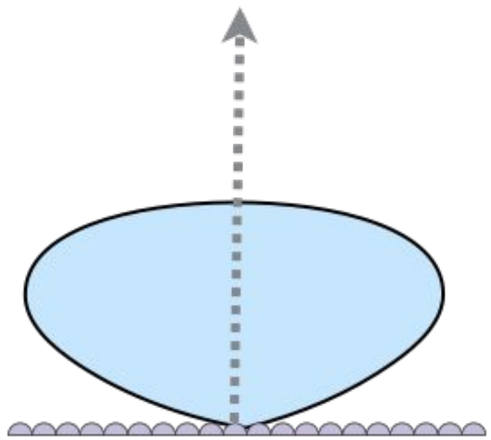
- Key: the distribution of microfacets' normals
  - Concentrated <==> glossy

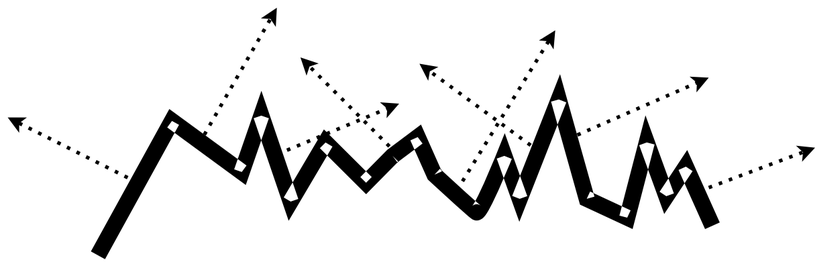


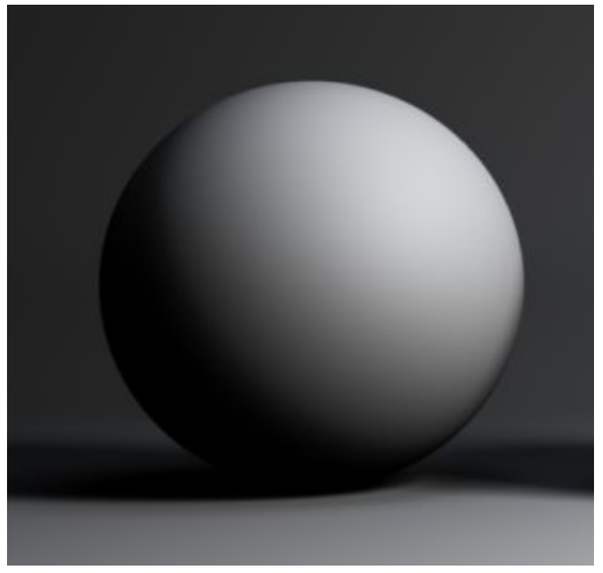




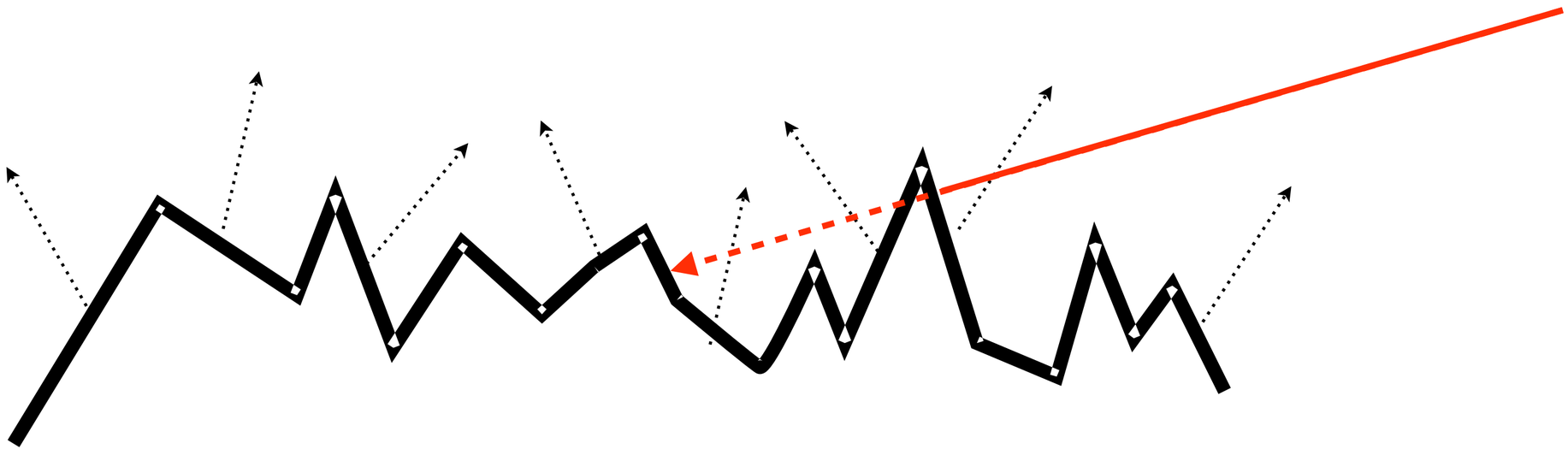
  - Spread <==> diffuse





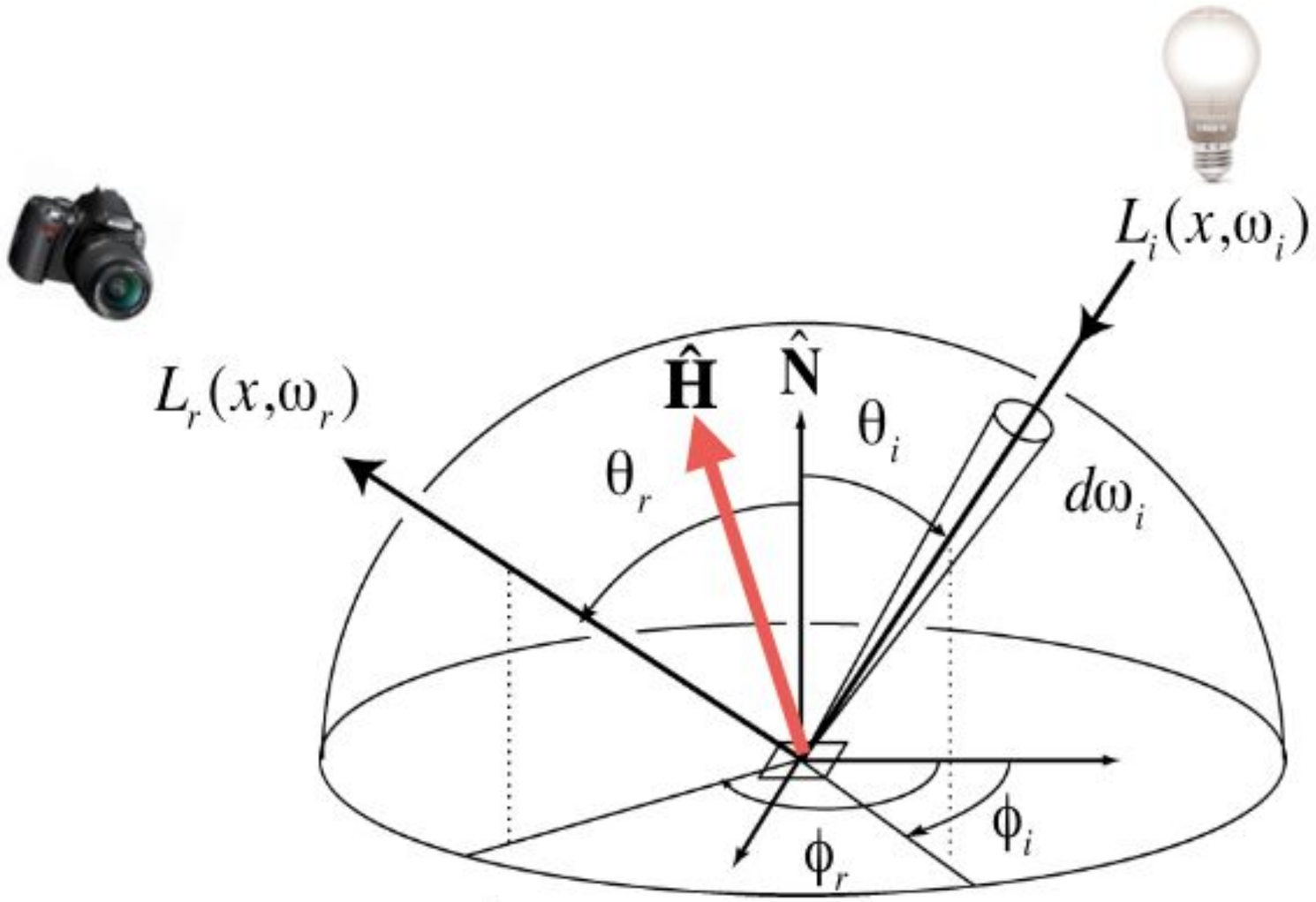


Shadowing/masking term



At grazing incoming light angles, some microfacets will block light from reaching other parts of surface

Important concept: the “half angle”



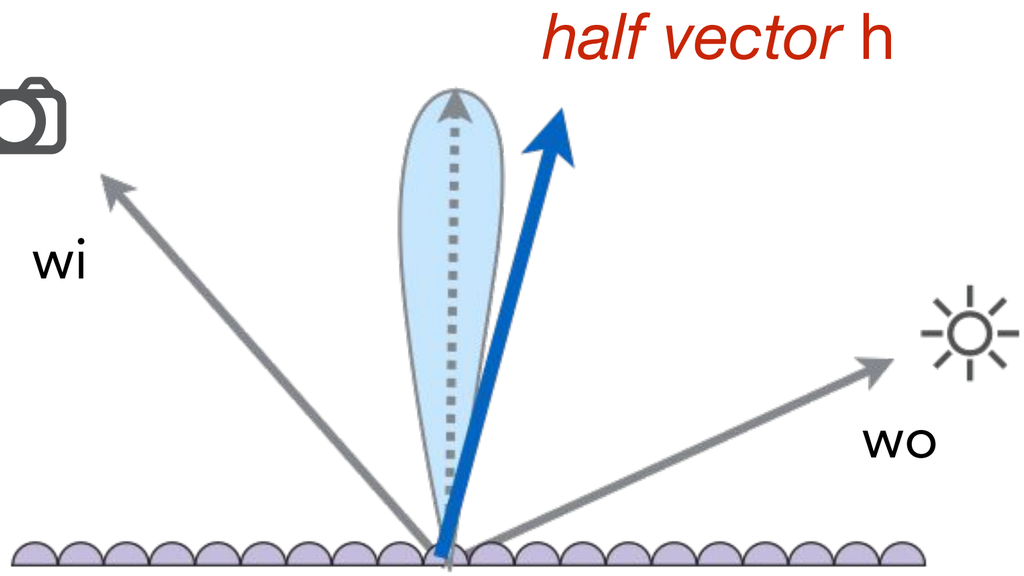
$$\hat{\mathbf{H}} = \text{normalize}(\omega_r + \omega_i)$$

Perfect specular reflection occurs when $\hat{\mathbf{H}} = \hat{\mathbf{N}}$

Glossy BRDFs often written as a function of $\hat{\mathbf{H}} \cdot \hat{\mathbf{N}}$

Microfacet BRDF

- What kind of microfacets reflect wi to wo? (hint: microfacets are mirrors)





Fresnel term

shadowing-masking term

distribution of normals

$$f(\mathbf{i},\mathbf{o}) = \frac{\mathbf{F}(\mathbf{i},\mathbf{h})\mathbf{G}(\mathbf{i},\mathbf{o},\mathbf{h})\mathbf{D}(\mathbf{h})}{4(\mathbf{n},\mathbf{i})(\mathbf{n},\mathbf{o})}$$

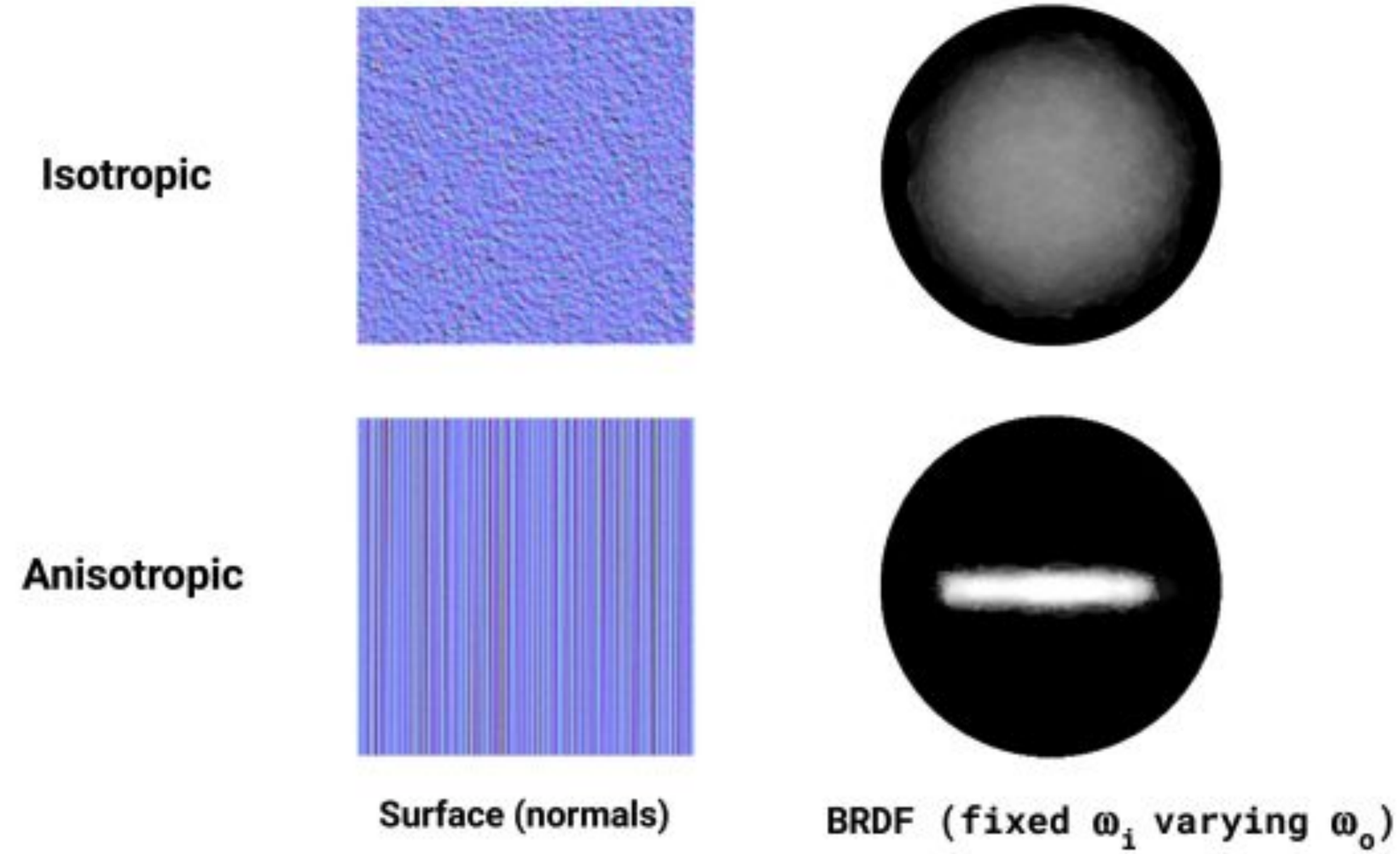
Isotropic / Anisotropic Materials (BRDFs)

Isotropic materials look the same regardless of how they are rotated around the surface normal. Their microstructure is random.

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) = f_r(\theta_i, \theta_r, \phi_r - \phi_i)$$

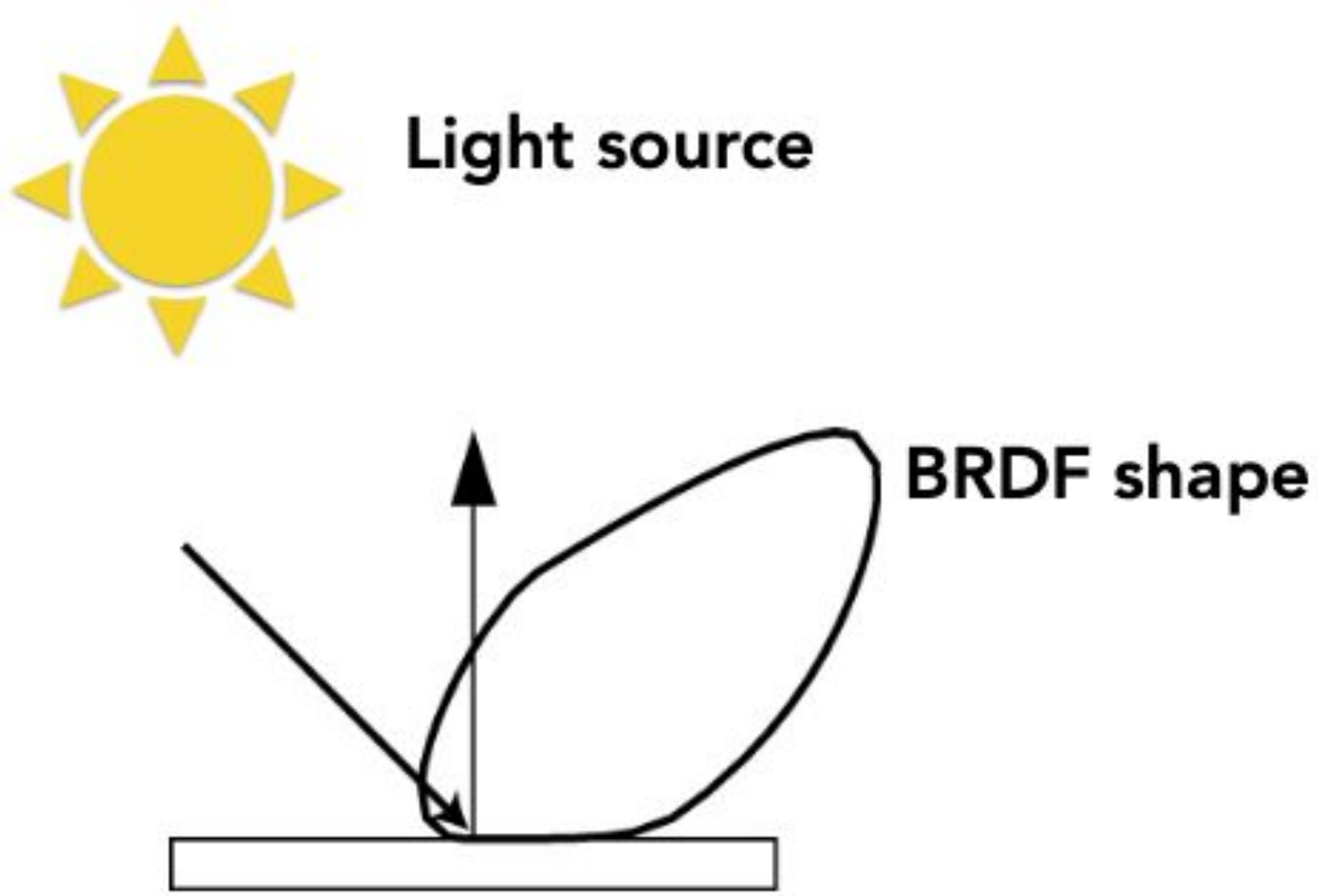
Anisotropic materials have an oriented microstructure, like brushed metal or wood grain. This causes reflections to stretch.

$$f_r(\theta_i, \phi_i; \theta_r, \phi_r) \neq f_r(\theta_i, \theta_r, \phi_r - \phi_i)$$



Importance Sampling

Importance sampling: lights and BRDF



$$L_r(\mathrm{p}, \omega_r) = \int_{H^2} f_r(\mathrm{p}, \omega_i \rightarrow \omega_r)\, L_i(\mathrm{p}, \omega_i)\, \cos\theta_i \, \mathrm{d}\omega_i$$

Importance sampling diffuse BRDF

$$L_o(\omega_o) = \frac{\rho}{\pi} \int L_i(\omega_i) \cos\theta_i \mathrm{d}\omega_i$$

The BRDF factors out of the reflectance integral since it's constant

Can just use cosine-weighted random samples on hemisphere

Importance sampling microfacet BRDF

distribution of normals

$$f(\mathbf{i},\mathbf{o}) = \frac{\mathbf{F}(\mathbf{i},\mathbf{h})\mathbf{G}(\mathbf{i},\mathbf{o},\mathbf{h})\mathbf{D}(\mathbf{h})}{4(\mathbf{n},\mathbf{i})(\mathbf{n},\mathbf{o})}$$

They come with a probability distribution built right in!

Sampling a half-angle from $\mathbf{D}$ works well to match specular lobe

Work Problem 2: Cook-Torrance Calculation

Worksheet Section 3, Problem 3 - ~10 minutes

3. **Walkthrough: Cook-Torrance BRDF Calculation**

   This problem will walk you through a full shading calculation for a single point on a surface using the Cook-Torrance microfacet model.

   Scenario:

   - A point on a surface has normal $\mathbf{n} = (0, 0, 1)$.
   - The viewer is in direction $\omega_o = (\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$.
   - A single point light provides radiance L_i from direction $\omega_i = (-\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$.
   - Assume for this setup, the material properties evaluate to $D = 1/\pi$, $G = 1$, and $F = F_0$.

   Your task is to calculate the final outgoing radiance, $L_o(\omega_o)$.

   1. Calculate the halfway vector, $\mathbf{h}$.
   2. Calculate the dot products $(\mathbf{n} \cdot \omega_i)$ and $(\mathbf{n} \cdot \omega_o)$.
   3. Assemble the full Cook-Torrance BRDF value, $f_r = \frac{D \cdot G \cdot F}{4(\mathbf{n} \cdot \omega_i)(\mathbf{n} \cdot \omega_o)}$.

   4. Calculate the final outgoing radiance L_o using the reflection equation for a single point light: $L_o = f_r \cdot L_i \cdot (\mathbf{n} \cdot \omega_i)$.

Solution 2: Halfway Vector & Dot Products

Worksheet Section 3, Problem 3

Solution:

1. **Halfway Vector h:**

$$\mathbf{h} = \frac{\omega_i + \omega_o}{||\omega_i + \omega_o||} = \frac{(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}) + (\frac{\sqrt{3}}{2}, 0, \frac{1}{2})}{||\dots||} = \frac{(0,0,1)}{||(0,0,1)||} = (0,0,1)$$

   The halfway vector is aligned with the normal, representing a perfect specular reflection condition.

2. **Dot Products:**

$$(\mathbf{n} \cdot \omega_i) = (0,0,1) \cdot (-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}) = \frac{1}{2}$$

$$(\mathbf{n} \cdot \omega_o) = (0,0,1) \cdot (\frac{\sqrt{3}}{2}, 0, \frac{1}{2}) = \frac{1}{2}$$

Solution 2: BRDF & Outgoing Radiance

Worksheet Section 3, Problem 3

3. **Assemble the BRDF f_r:**

$$f_r = \frac{D \cdot G \cdot F}{4(\mathbf{n} \cdot \omega_i)(\mathbf{n} \cdot \omega_o)} = \frac{(1/\pi) \cdot 1 \cdot F_0}{4 \cdot (1/2) \cdot (1/2)} = \frac{F_0/\pi}{1} = \frac{F_0}{\pi}$$

4. **Final Outgoing Radiance L_o:**

$$L_o = f_r \cdot L_i \cdot (\mathbf{n} \cdot \omega_i) = \frac{F_0}{\pi} \cdot L_i \cdot \frac{1}{2} = \frac{F_0 L_i}{2\pi}$$